

Students' Dropout and Academic Success

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Abstract—Student dropout and academic success are current challenges in higher education. This study analyzes a comprehensive dataset from a Portuguese institution, employing exploratory data analysis, dimensionality reduction, and clustering to uncover key patterns. We evaluate three models—MLP, Logistic Regression, and LightGBM—to predict academic outcomes, demonstrating that addressing data imbalance and leveraging machine learning can provide actionable insights for educational interventions.

Keywords— *academic success, educational data analysis, classification, data imbalance, oversampling techniques*

I. INTRODUCTION

Student dropout and academic success are concerns in educational institutions worldwide. Understanding the factors that influence student performance can help universities and policymakers develop strategies to improve retention rates and support academic achievement.

In this study, we analyze a dataset related to student academic performance, sourced from the UCI Machine Learning Repository. The dataset includes various demographic, socio-economic, and academic-related attributes collected at the time of enrollment. Our objective is to apply machine learning algorithms to predict student outcomes, specifically identifying students at risk of dropping out or underperforming.

This research is particularly relevant given the increasing availability of student-related data and the potential for data-driven decision-making in education. By using supervised learning techniques, we aim to classify students into different academic success categories, providing insights that could be leveraged by educational institutions to enhance student support programs.

A. Dataset Presentation

The dataset used in this study, titled Predict Students Dropout and Academic Success, is sourced from the UCI Machine Learning Repository. It consists of student records collected at a higher education institution in Portugal and contains a range of attributes related to student demographics (Age at enrollment, gender, nationality, and marital status), academic background (Admission scores, high school retention years, course choice preference, and prior academic performance), and socio-economic status ((Parents' educational background, employment status, and student financial aid). The dataset's primary goal is to predict student performance outcomes, making it suitable for both classification and regression tasks.

Its unique structure allows us to explore a multi-class classification problem that has real-world applications. Unlike traditional binary classification problems that predict success or failure, this dataset provides a nuanced view of student outcomes, incorporating a middle category that accounts for delayed graduation. This added complexity makes the dataset

particularly valuable for developing machine learning models that can differentiate between varying levels of academic risk.

B. Methodology

Our approach begins with an extensive exploratory data analysis to understand the intrinsic structure of the dataset, assess relationships among features, and identify any potential issues such as class imbalance. We then employ advanced analytical techniques including correlation analysis, dimensionality reduction via Principal Component Analysis (PCA), and clustering to uncover underlying patterns.

Finally, we apply supervised learning models—namely a Multi-Layer Perceptron (MLP), Logistic Regression, and LightGBM—to classify students into the predefined outcome categories. This multi-algorithm approach allows us to capture both linear and non-linear relationships in the data, providing robust insights that can guide targeted interventions and policy decisions in the educational sector.

C. State of the art

The field of research in education has been a subject of a wide variety of machine learning projects worldwide. Different variables such as demographic, socio-economic, and academic data were considered for the construction of these machine learning models. Beaulac and Roosenthal (2021)[1] achieved a prediction accuracy of 79% using Random Forests (RF) on a large dataset from a Canadian university, while Hoffait and Schyns (2020)[2] focused on identifying students likely to struggle during their first year, with mixed results across different algorithms.

The dataset studied in this article was already covered by another study carried on by Miguéis et al. (2020)[3] to predict academic performance using various machine learning models, resulting in an elevated accuracy with Random Forests and Adaptive Boosting Decision Trees. A very recurrent challenge of this area is dealing with imbalanced datasets because usually students at risk of failure are a minority in most of the cases. Thammasiri et al. (2020)[4] addressed this issue by employing synthetic oversampling techniques like SMOTE to balance classes, achieving better results than with standard methods.

Recent studies also highlight the importance of addressing imbalanced datasets, particularly for identifying minority classes like students at risk of failure, and demonstrate that boosting algorithms like XGBoost and LightGBM often outperform standard classification models. Despite improvements in prediction, challenges remain, especially in accurately identifying students in the "relative success" category, where intervention strategies are most needed. This study follows similar trends but emphasizes the use of different machine learning algorithms and imbalancing methods to compare the uncovered results based on the European dataset.

II. DATA ANALYSIS

A. General overview of the dataset

Before delving in the development of the models to achieve the proposed goals, an important step is to analyze the dataset as it is. This step, known as exploratory data analysis, helps us gain a deeper understanding of the actual state of the data and even draw some conclusions based on the relationships between the dataset's columns and the target column or even among the feature columns themselves.

Since the data originates from another study, the data cleaning step of exploratory data analysis was not necessary. The dataset used for further studies had already been cleaned, meaning that in its initial state, out of the 4,424 rows, there were no missing or null values, and no duplicate rows were present. Also, there is no significant outlier because the columns with continuous data have a fixed/specific range of values.

In total, the dataset contains 36 features or variables of different types. Generally, five types of variables can be found in the dataset:

- Continuous: Numbers that can take any value within a range
- Binary: Variables that only have two states (yes/no, 1/0, true/false).
- Discrete: Numbers that come from a countable set, usually whole numbers.
- Ordinal: Categories with a meaningful order, but differences between them are not measurable or uniform.
- Categorical: Variables that can take on one of a limited, fixed number of possible values, each representing a category or group.

The continuous columns contain the following information: Previous qualification (grade), Admission grade, Unemployment rate, Inflation rate, GDP, Curricular units 1st semester (grade), and Curricular units 2nd semester (grade). Although this classification does not influence the correlation study of the features, we can further divide the continuous columns into two groups: curricular information (grades) and socioeconomic-related data (inflation rate, GDP, and unemployment rate).

Regarding the binary data, the following columns are included: Daytime/evening attendance, Displaced, Educational special needs, Debtor, Tuition fees up to date, Gender, Scholarship holder, and International. These columns contain information about the student's characteristics and conditions throughout their college education.

In the categorical data group, we have: Application mode, Course, Previous qualification, Nationality, Target, Father's occupation, and Mother's occupation. These features provide background information about the student and their parents, the student's course completion status (Target variable), and the application method used for enrollment in higher education.

Finally, the ordinal features are found in the following columns: Marital status, Application order, Father's qualification, and Mother's qualification. While Marital status and Application order have an inherent order in their encoding, Father's and Mother's qualification do not. This presents two options for handling these features: either treat them as categorical variables or create an ordinal grouping.

The first approach is more straightforward but does not add additional meaning to these features. On the other hand, the second approach requires more data manipulation, which, if not carried out properly, could introduce errors. However, it can also enrich the information in these columns. If the original mapping is tracked, no information will be lost.

B. Correlations

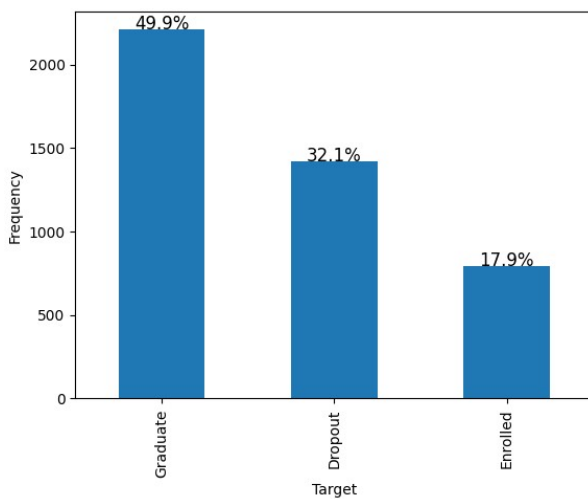
In this analysis, we confronted a dataset that includes continuous, binary, ordinal, and discrete variables. Given that the continuous features failed the Shapiro-Wilk normality test, traditional parametric methods like ANOVA were not appropriate. Instead, we binned the continuous variables into quartiles, which allowed us to transform them into categorical data and use Cramér's V—a robust measure for assessing associations between categorical variables. For the binary variables, we directly applied Cramér's V, which is well-suited for measuring relationships between nominal features and our target variable. The ordinal variables were examined using several complementary methods, including chi-square tests for trend, Spearman's rank correlation, and ordinal logistic regression, alongside a categorical treatment via Cramér's V. This multi-faceted approach was designed to capture both monotonic and categorical relationships, ensuring that the true associations were identified without relying on assumptions of normality.

For the discrete variables—such as those representing various counts related to curricular units—we employed a different strategy. First, we computed a correlation matrix and visualized it with a heatmap to reveal the linear relationships between these count variables. Next, pairwise Chi-Square tests provided overwhelming evidence of statistically significant associations across almost all discrete features. For example, the test between 'Curricular units 1st sem (credited)' and 'Curricular units 1st sem (approved)' yielded a very high chi-square statistic with a p-value effectively zero, confirming a strong association. Furthermore, we implemented a Poisson regression model to predict 'Curricular units 1st sem (approved)' using other discrete predictors. The model's strong fit, as evidenced by a high pseudo R^2 value, reinforced the conclusion that these variables significantly explain the variation in approved counts. Visualizations such as count plots, histograms, boxplots, and pairplots further illuminated the distributions and pairwise relationships among the discrete variables.

Overall, the significant findings from our analysis illustrate a consistent pattern: while some associations among binned continuous and binary variables are weak to moderate, the discrete data reveal robust interdependencies. The chi-square tests consistently produced highly significant p-values, and the Poisson regression model highlighted substantial contributions from several predictors. These results, supported by rigorous statistical testing and clear visual evidence, confirm that the observed associations are not due to random chance but reflect meaningful relationships within the data.

C. Advanced analysis of the dataset

When we take a closer look at the feature containing the classes we want to use for classifying new data, specifically the target column, we can see that the class distribution is not well balanced. Plotting a frequency-per-class chart can help us visualize an important aspect of our dataset: the imbalance of the mapped classes in the given data.



While we have a large proportion of graduates (Success - 49.9%) and a considerable number of dropouts (Failure - 32.1%) in the target feature, the enrolled (Partial Success - 17.9%) class is underrepresented. This imbalance can introduce significant challenges for the models we train on this data.

In this case, the imbalance is particularly problematic because one of the main goals of these models is to predict when a student is likely to face difficulties during their time in higher education—in other words, when they are at risk of being classified as partial success. The dataset's imbalance means that an algorithm trained on this data is more likely to develop a bias toward classifying students as graduates or dropouts, making it less sensitive to detecting cases of partial success. This directly affects the model's reliability and effectiveness in predicting student outcomes.

Once the data imbalance is identified, various strategies can be used to address it. Generally, these techniques are applied after splitting the data into training and test sets and completing preprocessing to avoid altering the dataset's original statistical properties.

For large datasets, undersampling may be ideal, as it reduces the number of observations from the majority classes while keeping all instances of the minority class. This ensures a balanced representation in the training set without artificially generating new data. For smaller datasets, where data loss is not an option, oversampling strategies are recommended. These methods artificially generate additional data for underrepresented classes using various techniques to improve class balance without discarding valuable observations.

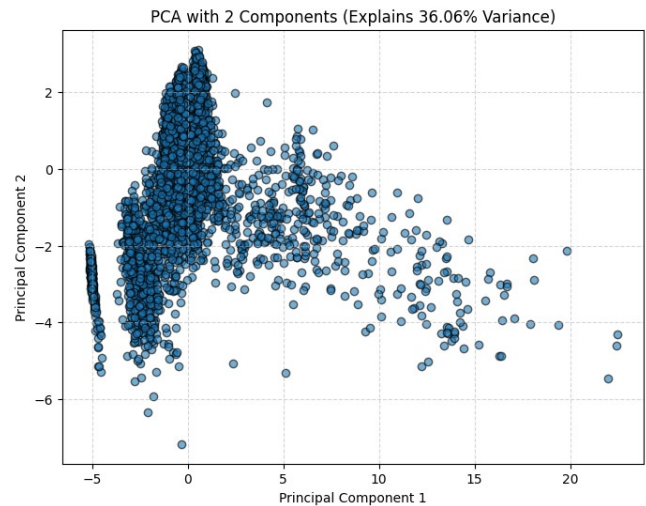
After performing traditional exploratory data analysis, we can enhance our insights further by using unsupervised machine learning algorithms to gain a broader perspective on the dataset. For this step, two techniques were applied: Principal Component Analysis (PCA) and clustering.

The first unsupervised method used was PCA, as it provides a transformed dataset that can benefit the subsequent clustering methods. In general, PCA is a linear dimensionality reduction technique that transforms the data into a new coordinate system, where the directions of maximum variance are more easily identifiable.

PCA was tested with two different approaches:

- Reducing the dataset's dimensionality to two principal components for visualization purposes.
- Retaining only the features that explain 95% of the dataset's variance.

In the first test, the output was visualized using a scatter plot. However, this representation was not ideal, as the two principal components only accounted for 36.01% of the dataset's variance.



The second test could not be represented visually in a plot, as the dataset's dimensionality was reduced from 36 to 21 dimensions. However, instead of using the original dataset, this reduced version was applied to the subsequent techniques.

Clustering was performed using three different algorithms:

- K-Means: An iterative, centroid-based clustering algorithm that partitions the dataset into k clusters based on the distance between data points and their respective centroids.
- DBSCAN: A density-based clustering algorithm that groups data points closely packed together while marking outliers as noise, based on the density of points in the feature space.
- GMM (Gaussian Mixture Model): A probabilistic model that assumes data points are generated from a mixture of Gaussian distributions with unknown parameters.

These algorithms were evaluated using the following metrics:

- Silhouette Score: Measures how well each data point fits within its assigned cluster. The score ranges from -1 to 1, with higher values indicating better-defined clusters. The final silhouette score is the average across all data points.
- Calinski-Harabasz Index: Measures the ratio of dispersion between clusters to dispersion within clusters. A higher value indicates well-separated clusters with low intra-cluster dispersion.

To optimize clustering performance, different hyperparameters were tested for each algorithm:

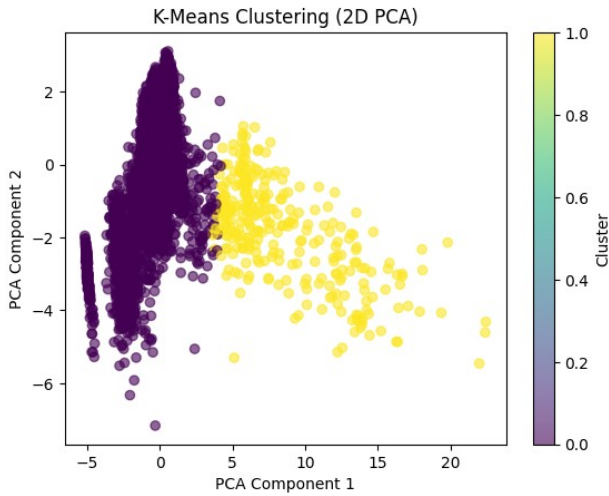
- K-Means: Iterated over different k values, selecting the one that produced the highest silhouette score.
- DBSCAN: Tested different epsilon (ϵ) values, choosing the one that yielded the highest silhouette score.

- GMM: Experimented with different numbers of components to find the best Bayesian Information Criterion (BIC) score.

The results of these tests were as follows:

Algorithm	Silhouette	Calinski-Harabasz Index	Ideal hyper-parameter
K-means	0,4484	990,89	K = 2
DBSCAN	-0,1356	42,62	Eps = 2.0
GMM	0,0496	254,03	N = 9

By interpreting the results of the analysis we can consider K-means with 2 clusters the ideal clustering algorithm, between the three tested, to apply to our dataset since it presents the highest silhouette score and highest Calinski-Harabasz Index. This clustering can be seen in the plot below.



III. MACHINE LEARNING PART

A. Preprocessing of the data

When developing a machine learning model, a variety of factors must be taken into consideration, such as the goal of the model, the current state of the dataset, the ideal data split, the choice of algorithm, and many others. However, before feeding data into the model to observe its outputs and performance, some data manipulation is necessary to format the input into a more algorithm-friendly structure, ensuring the model functions properly.

This stage of machine learning is called preprocessing, and it is a crucial step to ensure that the model is working with well-structured data. One critical aspect of this step is avoiding a major issue known as data leakage.

In simple terms, data leakage occurs when a model gains access to information from the test data before actually being tested, allowing it to "cheat" by leveraging knowledge of cases it has not yet encountered. Ideally, when testing, the model should only be exposed to new data that is different from the cases it interacted with during the training phase.

To prevent data leakage, certain measures can be taken. The first is splitting the dataset into training and test subsets before performing any preprocessing. This ensures that no

underlying information is transferred from the test set to the training set. For example, if data normalization is applied before the split, information about the range and values of the test set could leak into the training set, introducing bias and compromising the integrity of model training.

After splitting the data, necessary preprocessing steps can be carried out. These steps should consider the different data types in the dataset and the required format for the chosen algorithm.

The treatment applied to the training dataset depends on the selected algorithm. When dealing with quantitative data (discrete or continuous), normalization can be applied to prevent issues where certain features have vastly different scales, which could introduce bias into the algorithm. Additionally, when normalizing or standardizing quantitative data, the reference values should be derived solely from the training subset to avoid data leakage.

For qualitative data, encoding is often required since some algorithms are not designed to process textual data. If the features contain ordinal data, their encoded values should reflect their natural order to preserve meaningful relationships.

Furthermore, if the algorithm cannot handle textual data natively (such as k-NN or MLP), one-hot encoding may be necessary. This process transforms categorical variables into binary columns, ensuring proper processing. However, improper one-hot encoding can also introduce data leakage if different encoding parameters are applied to the training and test sets. Besides data leakage, inconsistent encoding can lead to feature dimension mismatches, causing errors during prediction.

B. Algorithm 1 – Multi-Layer Perceptron

The first algorithm selected for further analysis of the dataset was a Multi-Layer Perceptron, as it is a highly suitable classification algorithm that was not covered in the previous study on this dataset. Additionally, this algorithm can benefit significantly from a well-structured input dataset like ours and has the ability to learn complex patterns, such as non-linear relationships.

Despite its positive aspects, this algorithm also presents some challenges. It is prone to overfitting if the dataset is not large enough, is computationally expensive, and is difficult to interpret, often being considered a "black box."

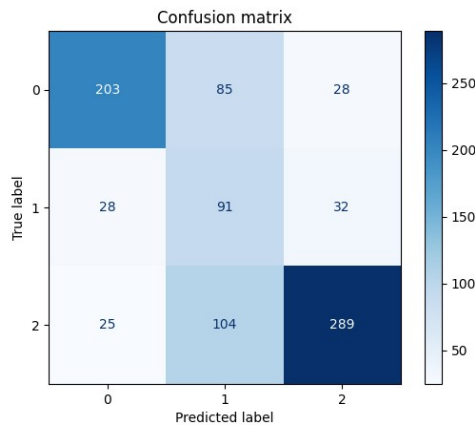
To evaluate the model's performance, the F1 score was used, as it considers both precision (the proportion of predicted positive cases that are actually correct) and recall (the proportion of actual positive cases that were correctly predicted). This metric is especially useful for imbalanced datasets, where high precision can be misleading.

Before optimizing the model's parameters, the performance of the algorithm using different oversampling techniques was evaluated. The selected oversampling techniques were Borderline-SMOTE, random over-sampling, SMOTE-ENN, and SMOTENC. These methods were chosen because they were not covered in the previous study on the same dataset. The following output was produced:

	No oversampling	Borderline-SMOTE	Random oversampling	SMOTE-ENN	SMOTENC
F1-score Failure	0.73	0.74	0.74	0.72	0.75
F1-score Rel. Success	0.39	0.40	0.41	0.42	0.41
F1-score Success	0.8	0.81	0.81	0.75	0.81
Average F1-score	0.64	0.65	0.66	0.63	0.66
Accuracy	0.70	0.71	0.72	0.66	0.73

Considering that performance in classifying the "Relative Success" class can add significant value to the model, the oversampling technique selected for parameter optimization was SMOTE-ENN. For parameter optimization, the chosen parameters were: hidden layer size, activation function, optimization algorithm, weight decay, and learning rate. These parameters were selected because they can directly impact accuracy and convergence while helping to prevent underfitting and overfitting.

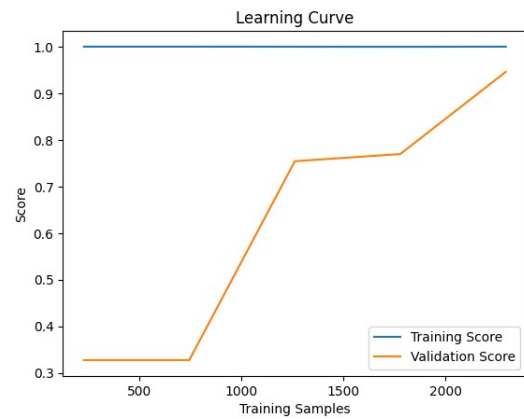
After applying hyperparameter optimization with the SMOTE-ENN method, the resulting accuracy was 0.94. One trade-off that is particularly evident when working with MLPs is the loss of explainability in exchange for accuracy. Although a high accuracy was achieved, the reasoning behind it remains unclear.



By analyzing the confusion matrix after optimization, we can observe that class 1 (Partial Success) has the lowest correct prediction rate, which is expected since it is the most challenging class to identify. Additionally, class 1 exhibits a high misclassification rate, while class 2 (Dropout) shows a high false positive rate.

Finally, by examining the learning curve of the model, we can conclude that the model is overfitting. The training score is 1.0 (the maximum value), indicating that the model has memorized the training data entirely. Furthermore, the validation score starts off very low but improves with more data. This suggests that the model generalizes poorly with

smaller training sets, which is a typical characteristic of this type of model.



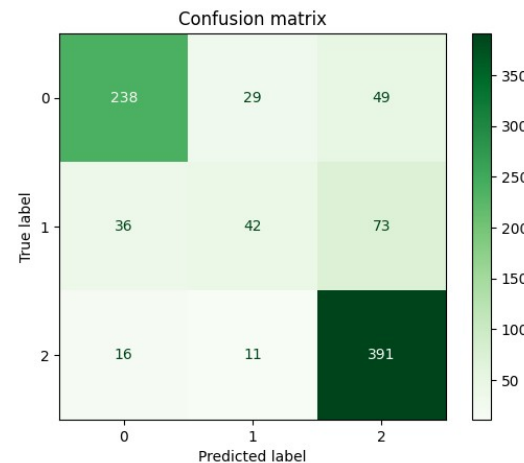
C. Algorithm 2 – Logistic Regression

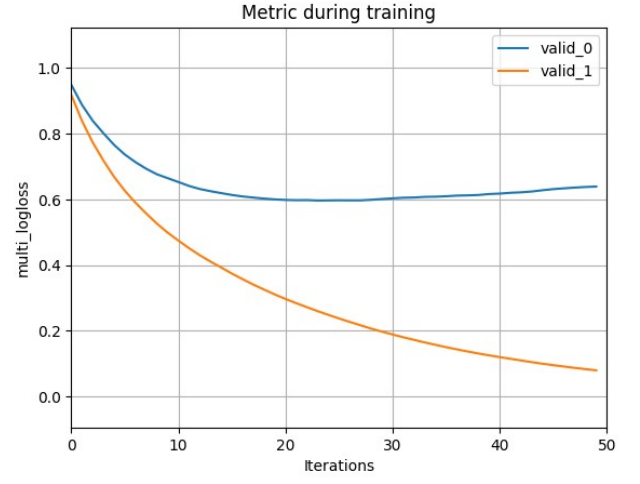
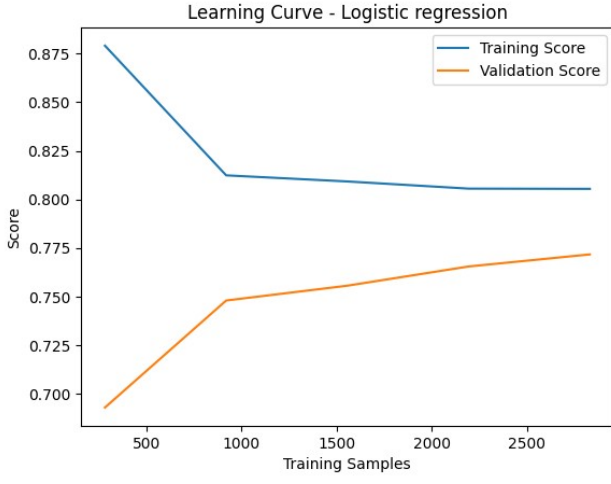
We chose Logistic Regression in order to establish a strong baseline. As a generalised linear model, it provides clear insights into the impact of different features on classification but is limited in capturing complex, non-linear patterns. The dataset's class imbalance is particularly problematic, as one of the main goals of this model is to identify students at risk of partial success. However, Logistic Regression's bias toward classifying students as graduates or dropouts makes it less sensitive to these cases, directly impacting its reliability in predicting student difficulties.

Random Search was used to optimize regularization strength (C), solver type, and penalty function, aiming to improve model generalization and avoid underfitting. The best parameters selected were 'saga' solver, the 'l2' penalty and C as a float of (1.623776739188721). This led to an increased model performance accuracy of 0.77 from 0.75 while preventing excessive variance.

Feature importance was evaluated using coefficient magnitudes with Curricular units in the 2nd semester having the most importance.

The confusion matrix indicated that Logistic Regression struggled with class 1 (Enrolled), achieving only a recall of 0.28, suggesting underfitting. The learning curve showed that the model generalized well but lacked the complexity needed to capture more nuanced relationships in the data making it insufficient on its own for identifying students at risk.





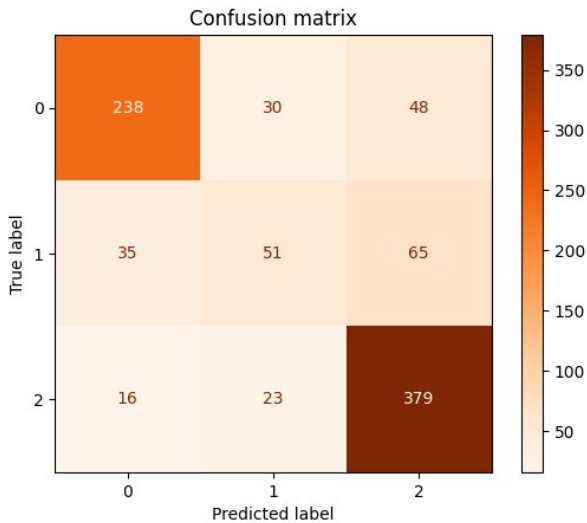
D. Algorithm 3 - LightGBM

We chose LightGBM for its ability to handle non-linear relationships and optimize decision boundaries through boosting. Unlike Logistic Regression, it can identify complex feature interactions which made it a strong choice for this particular dataset. However, class imbalance remains a challenge, as the dataset naturally favors predicting graduates and dropouts over enrolled students. Since one of the main goals is to detect students at risk of partial success, LightGBM's bias toward majority classes still affects its reliability in this area.

We used random search as well to finetune the parameter as we were prioritizing speed. The parameters we fine-tuned are: num_leaves, learning rate and boosting iterations.

From the feature evaluation, it is interesting to note that the feature that was most important was the Admission Grade.

The confusion matrix showed that LightGBM outperformed Logistic Regression in classifying class 1 (Enrolled) but still exhibited some miscalculations. It still struggles with class imbalance, making additional methods such as cost-sensitive learning or adjusted class weights necessary for improving class 1 recall. The learning curve revealed potential overfitting. This means that we still have to do further regularisation.



E. Comparison of the algorithms

Model	F1-Score (Dropout)	F1-Score (Enrolled)	F1-Score (Graduate)	Average F1-Score	Accuracy
MLP (plain)	0.76	0.39	0.80	0.64	0.70
Logistic Regression	0.79	0.36	0.84	0.66	0.76
LightGBM	0.79	0.40	0.83	0.67	0.75

Logistic Regression remains the fastest and most interpretable model due to its simple, linear nature [5]. However, it struggles with non-linearity and underfitting, particularly in classifying "Enrolled" students (recall: 0.36). LightGBM offers a better balance between complexity and efficiency, outperforming Logistic Regression in classifying class 1 (F1-score: 0.40 vs. 0.36) while maintaining moderate interpretability through feature importance analysis [6]. However, it still struggles with class imbalance despite oversampling and shows signs of slight overfitting, requiring further tuning. MLP achieved the highest recall for class 1 (0.39) due to its ability to capture complex patterns, but it remains computationally expensive and prone to overfitting [7].

Although oversampling techniques (SMOTE-ENN, Borderline-SMOTE, Random Oversampling) improved class 1 recall, further optimization is needed to determine the best approach. Feature importance for both Logistic Regression and LightGBM highlighted key variables affecting classification, and a deeper analysis of overfitting and underfitting would further refine hyperparameter tuning.

IV. CONCLUSION

This study tackled the pressing issue of predicting student dropout and academic success by analyzing a rich dataset from a Portuguese higher education institution. Through extensive exploratory data analysis, dimensionality reduction via PCA, and clustering, we identified underlying patterns that informed the subsequent application of three supervised learning models—MLP, Logistic Regression, and LightGBM. Our findings indicate that addressing data imbalance is key to accurately

classifying students, particularly in the underrepresented "Partial Success" category, and that each model offers unique insights into the factors affecting academic outcomes.

The selection of these machine learning methods was driven by their distinct advantages in addressing the complexities of educational data. The Multi-Layer Perceptron was employed for its strength in modeling intricate, non-linear relationships among diverse features, which is critical when dealing with the multifaceted nature of student performance. Logistic Regression served as a transparent baseline, providing clear insights into the influence of individual predictors and allowing us to understand linear trends within the data. Meanwhile, LightGBM was chosen for its efficiency and robust handling of non-linear interactions, making it well-suited for capturing subtle patterns that other methods might overlook. Together, these techniques complement each other, ensuring a comprehensive analysis that balances interpretability with predictive power.

In summary, this research demonstrates that machine learning can play a pivotal role in educational analytics by providing actionable insights to identify at-risk students and guide targeted interventions. The promising results not only validate our modeling approach but also suggest several avenues for future exploration. Potential next steps include integrating additional data sources, experimenting with ensemble and cost-sensitive learning methods to further

address class imbalance, and developing real-time predictive systems. These enhancements could lead to even more effective strategies for improving student retention and academic performance, ultimately contributing to more informed decision-making in the educational sector.

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